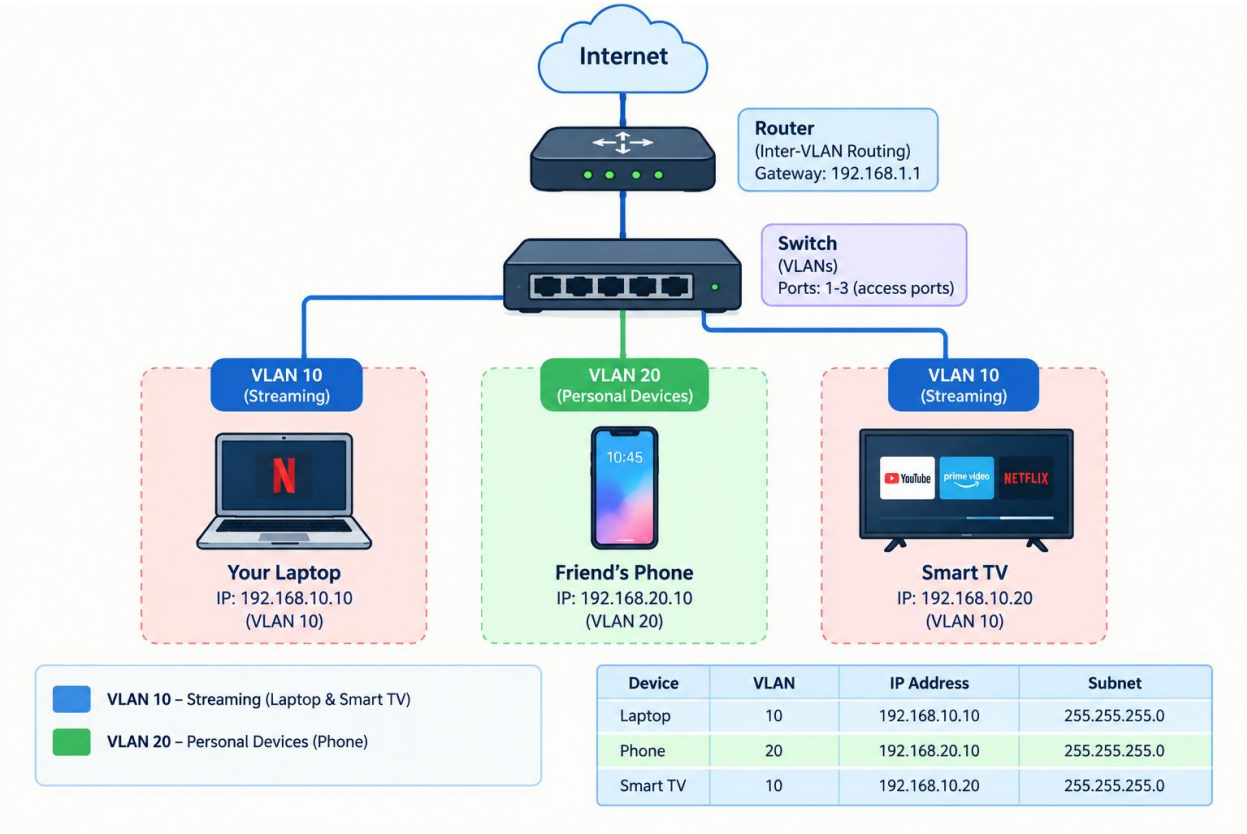
Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Device	VLAN	IP Address	Subnet
Laptop	10	192.168.10.10	255.255.255.0
Phone	20	192.168.20.10	255.255.255.0
Smart TV	10	192.168.10.20	255.255.255.0

**Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

VLAN Configuration in Cisco Packet Tracer

Aim: Create VLAN 10 and VLAN 20, connect five PCs to a switch, and test whether PCs in different VLANs can communicate.

1. Network diagram

Cisco Switch (2960)

VLAN 10 and VLAN 20

VLAN 10

PC1 · PC2 · PC3

192.168.10.1–3

VLAN 20

PC4 · PC5

192.168.20.1–2

Connect all five PCs to the switch using copper straight-through cables.

2. Assign IP addresses to the PCs

PC	IP address	Subnet mask	VLAN
PC1	192.168.10.1	255.255.255.0	10
PC2	192.168.10.2	255.255.255.0	10
PC3	192.168.10.3	255.255.255.0	10
PC4	192.168.20.1	255.255.255.0	20
PC5	192.168.20.2	255.255.255.0	20

In Packet Tracer, open each PC and go to Desktop → IP Configuration to enter the details. Leave the default gateway blank for this test.

3. Configure the switch

Open the switch, select CLI, and enter these commands:

```
enable
configure terminal

vlan 10
name VLAN10
exit

vlan 20
name VLAN20
exit

interface range fastethernet 0/1-3
switchport mode access
switchport access vlan 10
exit
```

```
interface range fastethernet 0/4-5
switchport mode access
switchport access vlan 20
exit

end
show vlan brief
```

What the commands do:

- `vlan 10` and `vlan 20` create the two VLANs.
- `name` gives each VLAN a name.
- `interface range` selects multiple switch ports.
- `switchport mode access` sets those ports as access ports.
- `switchport access vlan 10` or `20` assigns the ports to the correct VLAN.
- `show vlan brief` verifies the VLANs and port assignments.

4. Test the connection

Open PC1 → Desktop → Command Prompt and type:

```
ping 192.168.10.2
```

Expected result: Success, because PC1 and PC2 are in VLAN 10.

Now type:

```
ping 192.168.20.1
```

Expected result: Failure, because PC1 and PC4 are in different VLANs and no router or Layer 3 switch is configured to route traffic between them.

5. Final result

Test	Expected result	Reason
PC1 → PC2	Success	Same VLAN
PC1 → PC3	Success	Same VLAN
PC4 → PC5	Success	Same VLAN
PC1 → PC4	Failure	Different VLANs
PC2 → PC5	Failure	Different VLANs

Conclusion: VLANs divide a switch network into separate logical networks. PCs in the same VLAN can communicate, but PCs in different VLANs cannot communicate unless inter-VLAN routing is configured.

In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Moving a PC from VLAN 10 to VLAN 20 in Cisco Packet Tracer

Aim: Move PC3 from VLAN 10 to VLAN 20, then test connectivity using the `ping` command.

1. Network setup

Cisco Switch

PC1 → Fa0/1 · PC2 → Fa0/2 · PC3 → Fa0/3 · PC4 → Fa0/4

VLAN 10

PC1, PC2, PC3

192.168.10.0/24

VLAN 20

PC4

192.168.20.0/24

2. Before moving PC3

Enter these commands in the switch CLI:

```
enable
configure terminal
vlan 10
vlan 20
interface fastethernet 0/3
switchport mode access
switchport access vlan 10
end
show vlan brief
```

Explanation: PC3 is connected to port Fa0/3, which belongs to VLAN 10.

3. Test connectivity before moving

Set these IP addresses using PC → Desktop → IP Configuration:

PC	IP address	VLAN
PC2	192.168.10.2	10
PC3	192.168.10.3	10
PC4	192.168.20.4	20

On PC3, open Command Prompt and run:

```
ping 192.168.10.2
```

Expected result: Success, because PC2 and PC3 are in the same VLAN and subnet.

4. Move PC3 to VLAN 20

Enter these commands:

```
enable
configure terminal
interface fastethernet 0/3
switchport access vlan 20
end
show vlan brief
```

Explanation: The `switchport access vlan 20` command moves port Fa0/3 from VLAN 10 to VLAN 20. You do not need to create VLAN 20 again if it already exists.

For the test to work correctly, also change PC3's IP address to `192.168.20.3` with subnet mask `255.255.255.0`.

5. Test connectivity after moving

On PC3, run these commands:

```
ping 192.168.20.4
```

Expected result: Success, because PC3 and PC4 are now in VLAN 20 and the same subnet.

Now run:

```
ping 192.168.10.2
```

Expected result: Failure, because PC2 is in VLAN 10 and no inter-VLAN routing is configured.

6. Final result

Test	Before moving PC3	After moving PC3
PC3 → PC2 (VLAN 10)	Success	Failure
PC3 → PC4 (VLAN 20)	Failure	Success

Conclusion: Changing the switch port's access VLAN moves the PC into a different logical network. Its IP address must also match the new subnet for normal communication.

**Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.**

Cisco Packet Tracer: Assign 10 Ports to VLAN 30

We can use the Cisco IOS `interface range` command to assign ports FastEthernet 0/1 to 0/10 to VLAN 30 without repeating commands.

Commands

```
enable
configure terminal

vlan 30
name VLAN30
exit

interface range fastethernet 0/1-10
switchport mode access
switchport access vlan 30
exit

end
show vlan brief
```

Explanation in easy language

- `vlan 30` — Creates VLAN 30.
- `name VLAN30` — Gives the VLAN a name.
- `interface range fastethernet 0/1-10` — Selects all 10 ports at once.
- `switchport mode access` — Sets the ports to access mode.

- `switchport access vlan 30` — Assigns all 10 ports to VLAN 30.
- `show vlan brief` — Checks that the ports are assigned correctly.

Expected result: FastEthernet ports Fa0/1 through Fa0/10 should appear under VLAN 30.

Conclusion: The `interface range` command saves time because you can configure multiple switch ports together instead of configuring each port separately.